A PRELIMINARY OVERVIEW OF AN ALGORITHM FOR FINDING OPPOSITES (CONTRADICTIONS)

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Abstract. This is a preliminary overview of an algorithm for finding opposites (contradictions) and the mathematical expression of such an algorithm. Here we show only the main stages of the algorithm, as well as the main applicable tools.

TLDR

(1) The initial task was to learn how to determine opposites (contradictions) for the *papanda* application under development. Determining opposites is of key importance within dialectical analysis, and the papanda application is specifically devoted to such analysis.
(2) To find opposites (contradictions), one must choose the way in which the processes occurring in the world are described. There are three possible approaches to this: 1) laws, for example, the laws of physics; 2) probability; 3) dialectics. The first two solutions have their weaknesses from the standpoint of creating AI. The dialectical approach should, theoretically, have fewer limitations.
(3) If the first two approaches — laws and probability — represent the world as objects and the movement of those objects, dialectics represents the world as movement, of which objects may be a particular case.
(4) Dialectics works with the world as infinity, in which the simplest process develops, its opposite is found in this development, and the opposite turns into a contradiction.
(5) Mathematics (its main branches), and therefore AI, can determine only opposites, but cannot work with contradictions. Therefore, within the framework of the proposed algorithm, an opposite is taken as an indicator of a contradiction.
(6) In the course of searching for the opposite (contradiction), one can define an operator H that describes the development of the process. This operator, it is assumed, can serve as the basis for generation from already given information.

Some Physical Intuitions

To make what follows more intuitive, we propose using two images.

The first image. Imagine an infinite string of lights. At one moment, some of the lights are off and some are on. At another moment, different lights are off, while the remaining ones are on. Now imagine these moments replacing one another an infinite number of times. This creates a kind of "flickering" with a certain frequency. A numerical characteristic of such flickering, or the flickering itself, can express any feature, property, or action. This is possible only because we are dealing with infinity.

The second image. Imagine a Rubik's Cube. You are looking at one of its sides. At the next moment, the number of "small cubes" that make up the Rubik's Cube arbitrarily increases or decreases. There may be infinitely many such moments, and at each stage a change occurs. Through such change over infinity, we can obtain any figure; each figure will differ from another by a number and therefore can also express a feature, property, or action.

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The world can be understood in three ways. One can represent the world as a set of objects that move. In that case, one needs to know the laws governing the movement of those objects. This is what, for example, classical physics deals with. Another approach is related to probability. When it is impossible to find a law of motion, one can assign a probability to such motion.

The third approach is different from these two. There is no need to divide the world into objects and movement. Everything is treated as movement. You are not looking for the law according to which an object moves; you are looking for the law of movement itself.

Let us explain this using an example that partly resembles the well-known example associated with Einstein's theory of relativity. Take a sheet and stretch it. Throw a ball onto it. Under the first approach, you need to know the laws of elasticity, friction, and so on in order to determine where the ball will end up. Under the second approach, you can calculate the probability of the ball appearing at one point or another on the sheet. Under the third approach, there is neither a ball nor a sheet. There is movement.

Imagine that you recorded this on video: we stretch the sheet, throw the ball... In the video, you do not have the sheet or the ball as objects; you have only a sequence of frames, only movement. Here we may also say that there is "flickering" — this is why we needed the images above. Thus, you need to determine not the law governing the movement of the ball and not the probability of its appearing at a particular point, but the law according to which the frames of your video recording change.

It is important to note the following. We are not entering the field of physical science here and are not making claims about how the world is structured "in reality." What interests us is this: to obtain a picture of the world that can be worked with in such a way that it becomes possible to create an algorithm for finding opposites (contradictions) and generating.

Algorithm

The third approach described above can be expressed in stricter terminology as follows.

We represent the entire world as a process containing an infinite number of other processes, including processes nested within one another.

All processes develop. Development is understood as the emergence of a developing process from the simplest process, in which this developing process was already contained in potential form, while at the same time the simplest process itself is refined and disclosed through the emergence of a disclosing process.

The development of a process begins with the simplest process.

The simplest process is denoted as P_0. For our purposes, the simplest process is specified by the researcher.

The simplest process develops in an infinite world. Therefore, the complete set of developing processes cannot be reached.

In its infinity, the world is one; therefore, at infinity, any process is connected with any other process.

At the same time, the simplest process is considered at a specific iteration. Within a specific iteration, the simplest process is disclosed through any other processes.

An iteration means precisely the reduction of infinity to a finite set of developing processes.

The simplest process is disclosed through developing processes. Disclosure means the participation of a process in relations with other processes within a specific iteration.

Developing processes are denoted as P_1, P_2, and so on. For our experimental purposes, developing processes are generated by a neural network in accordance with the requirements imposed on developing processes by the present algorithm. The number of developing processes changes at each iteration, although the numbers may also coincide.

Each developing process, except the first one, follows from the simplest process and the preceding developing processes.

Each developing process contains a subsequent developing process within itself. This means that the development of the current developing process produces the subsequent developing process.

Each developing process has internal processes. To determine the internal processes, the developing process is taken as the simplest process, after which its developing processes are found without further nesting.

Internal processes are generated by a neural network in accordance with the requirements of the present algorithm. Their number changes between iterations, although coincidences in number are permitted.

Developing processes are compared with the simplest process. Their internal processes are compared with one another and with developing processes. On the basis of the results, levels, connection weights, and interactions between levels and weights are determined.

During the development of the simplest process, an opposite process is discovered among the developing processes.

An opposite process is a process that, in its development, excludes the existence of the simplest process. For the purposes of this algorithm, the detection of such opposition is taken as the criterion for detecting a contradiction.

Thus, starting from the simplest process, we obtain iterations in each of which the number of developing processes and internal processes increases or decreases. Metrics are constructed at each iteration.

Two neighboring iterations are regarded as the left-hand and right-hand sides of an equation. On each side of the equation is a record of the metrics of the corresponding iteration.

The left-hand and right-hand iterations cannot be equal because we are speaking about change from one iteration to another.

However, the left-hand and right-hand iterations are equal because we have agreed to represent the world as movement.

Movement is the simultaneity of the moments "here" and "there." We must no longer be "here," otherwise we are not moving, and we cannot yet be "there," otherwise we would not be moving.

Thus, the two sides of the equation are both equal and unequal.

How can this be achieved?

Through a hidden term of the equation.

We are accustomed to writing all terms of an equation either on the right-hand or on the left-hand side. In this case, however, the hidden term acts on the left-hand side of the equation, making it equal to the right-hand side at the moment when the left-hand side — the iteration — passes into the right-hand side — the next iteration.

In other words, whereas in an ordinary equation equality follows from the left-hand and right-hand sides, in our case the left-hand and right-hand sides follow from equality.

This hidden term determines the process by which iterations — frames, flickering — replace one another.

As a result, we have:

(1) an established contradiction in the development of the process;
(2) a hidden term that determines movement.

For this, we did not need to introduce the laws of physics, apply probability, or pretrain a separate generative model of the law of motion.

Mathematical Model

The mathematical exposition in this section was prepared by ChatGPT (OpenAI) on the basis of the author's ideas, problem formulation, and algorithm developed by Ruslan Nazarov. The concept of the method and the architecture of the algorithm were proposed by the author; ChatGPT's role consisted of formalizing and presenting the mathematical construction.

The purpose of the mathematical formalization is not to describe the world as a collection of predefined objects, but to construct a finite representation of a developing process and a mathematical description of the transition between such representations.

A computationally infinite process cannot be disclosed completely. Therefore, at each step only a finite configuration of processes and relations between them is considered. A sequence of such configurations forms the observable development of the original process.

1. Finite Representation of a Process. Let P_0 be the simplest process specified by the researcher.

At iteration k, a finite set of developing processes is considered:

$$P_k = \{P_i : i \in I_k\}$$

where I_k is the set of their indices.

The number of processes is not assumed to remain constant across different iterations:

$$|I_{k+1}| \neq |I_k|$$

in the general case. In other words, the next iteration may contain more, fewer, or the same number of developing processes as the preceding iteration.

Each developing process P_i corresponds to a finite set of internal processes:

$$U_i^{(t)} = \{U_{i1}, U_{i2}, \ldots, U_{im}\}$$

where t denotes the current step in the development of the observed picture.

If a process continues to exist in the history under consideration, its previously identified internal processes are preserved, while subsequent disclosure may expand their set:

$$U_i^{(t)} \subseteq U_i^{(t+1)}$$

Thus, change in the system consists not only in the appearance of new processes but also in the refinement of already existing ones.

At the same time, the absence of a process, an unknown value of a certain relation, and a numerically zero relation are treated as different states. Zero denotes a defined numerical value, not the absence of information.

2. Numerical Representation of Relations. To move from a substantive description of processes to a mathematical one, a semantic relation function is introduced:

$$s(a, b \mid C) \in [0, 1]$$

where C is the context in which the comparison is performed.

The value of s is not interpreted as the probability that a statement is true and, in the general case, is not a metric distance. It is a numerical characteristic of the substantive correspondence between two processes under consideration in a fixed context.

The relation between a developing process P_i and the original process is defined as:

$$\ell_i = s(P_i, P_0)$$

For an internal process:

$$A_{ia}^0 = s(U_{ia}, P_0)$$

Relations between internal processes belonging to different parents are also considered:

$$B_{ia,j\beta} = s(U_{ia}, U_{j\beta}), \qquad i \neq j$$

as well as the relation between an internal process of one developing process and another developing process:

$$C_{ia,j} = s(U_{ia}, P_j), \qquad i \neq j$$

Thus, a developing process is not characterized by a single number. It is embedded in a system of relations between the original process, other developing processes, and their internal structures.

3. Local Structure of Relations. Consider an internal process U_{ia} and another developing process P_j.

Let

$$c = C_{ia,j}$$

and let b denote the strongest relation between U_{ia} and the internal structure of P_j:

$$b = \max_{\beta} B_{ia,j\beta}$$

Thus, c characterizes the direct relation between the internal process and P_j, whereas b characterizes the strongest relation with its internal structure.

To represent these two relations jointly, we introduce four components:

$$Q(c,b) = \big(\min(c,b);\ (c-b)_+;\ (b-c)_+;\ 1-\max(c,b)\big)$$

where

$$(x)_+ = \max(x,0)$$

These four components are non-negative, and their sum equals one:

$$Q_1 + Q_2 + Q_3 + Q_4 = 1$$

They can be understood as a decomposition of a pair of relations into:

- a jointly represented part;
- the advantage of the direct relation;
- the advantage of the relation through the internal structure;
- a residual part.

At the same time, the use of $\min(c,b)$ is a mathematical convention for decomposing two numerical estimates. It does not in itself prove the semantic identity of the corresponding processes.

For a group of internal processes, local values may be aggregated. In simplified form:

$$W_{ij} = \frac{1}{|V_{ij}|} \sum_{a} Q_{ia,j}$$

where V_{ij} is the set of internal processes for which the necessary relations are defined.

As a result, W_{ij} characterizes not an individual comparison but the structure of the relation of one developing process to another through its internal disclosure.

Moreover, in the general case:

$$W_{ij} \neq W_{ji}$$

That is, the direction of the relation remains significant.

The complete mathematical construction contains additional derived indicators and interactions between levels and relations. They are not considered in detail in the present article. To understand the basic principle, the following transition is sufficient:

$$\text{semantic estimates} \ \rightarrow\ \text{local relations} \ \rightarrow\ \text{aggregated structure}$$

4. Mathematical "Frame" of the Process. Simply adding together the obtained numbers would destroy information about which processes and relations they belong to. Therefore, each quantity must retain its address.

Let $D_k^{(t)}$ be the set of addresses of the numerical characteristics of iteration k at step t.

The state of a single iteration can then be represented conditionally as:

$$X_k^{(t)} = \sum_{\alpha} x_{k,\alpha}^{(t)} \cdot e_{k,\alpha}$$

where α denotes the address of a characteristic and $e_{k,\alpha}$ denotes its corresponding position in the state structure.

The complete observable picture at step t is defined by the combination of the states of the available iterations:

$$Z_t = X_0^{(t)} \oplus X_1^{(t)} \oplus \cdots \oplus X_t^{(t)}$$

It is Z_t, rather than a separate set of average indicators, that is subsequently regarded as a finite mathematical "frame" of the developing process.

The size of such a frame does not have to remain constant. A new iteration may contain a different number of processes, while already existing processes may acquire additional internal disclosure.

Development therefore takes the general form:

$$Z_t \;\rightarrow\; Z_{t+1}$$

This is a transition between states that differ in composition but are structurally comparable.

5. The Motion Operator H. In the algorithmic part, two neighboring iterations were considered simultaneously as different states and as two moments of one movement.

Mathematically, this does not require us to claim:

$$Z_t = Z_{t+1}$$

On the contrary, in the general case:

$$Z_t \neq Z_{t+1}$$

The unity of the two states is expressed not by their identity, but by the presence of a common transition structure between them.

To describe such a structure, an operator H is introduced.

Let H_t denote the history of the system states available by step t.

For a certain numerical characteristic with address α, the transition can then be represented in the general form:

$$\eta_{t+1,\alpha} = \theta_1 \Phi_{1,\alpha} + \theta_2 \Phi_{2,\alpha} + \cdots + \theta_r \Phi_{r,\alpha}$$

Here, each function Φ depends on the accumulated history H_t and on that part of the next state which is permitted to be used in the corresponding calculation. More compactly:

$$\eta_{t+1,\alpha} = \sum_v \theta_v \cdot \Phi_{v,\alpha}(H_t, V_{t+1})$$

where Φ_v are functions that identify certain structures of movement, θ_v are common coefficients, and V_{t+1} is the permissible part of the information from the next state.

It is essential that a separate free coefficient is not introduced for each new process in a way that would automatically allow any two pictures to be reconciled. Otherwise, the operator H would simply become a record of the already observed difference:

$$Z_{t+1} - Z_t$$

and would cease to express the general character of movement.

The coefficients of H are determined from the accumulated history. In general terms, a set of coefficients Θ is sought that minimizes the reconciliation error while simultaneously limiting the complexity of the model:

$$\Theta^* = \arg\min \left[\text{weighted mean error} + \lambda \cdot \text{complexity of } \Theta \right]$$

where $\lambda > 0$ is the regularization parameter.

For characteristics that are meaningful only within the interval from 0 to 1, the resulting value is restricted to this range:

$$\hat{y} = \min(1, \max(0, \eta))$$

Thus, the law of transition is not given to the system by the researcher in a finished form. What is investigated is whether a common structure of transitions can be obtained from the history of observed states.

At the same time, H **is not used as a substantive generator of new developing processes**. The generation of new disclosure and the mathematical description of already obtained movement are different operations.

The specific dictionary of functions Φ, the complete set of permissible dependencies, and the precise procedure for constructing individual contractions are not provided in the present article.

6. History as a Trajectory. A single transition

$$Z_t \;\rightarrow\; Z_{t+1}$$

is insufficient to describe movement.

For a process P_i, its history is considered on a common time scale. Abstractly, it can be represented as a trajectory:

$$Y_i : [0,1] \to \mathbb{R}^d$$

The coordinates of Y_i describe the appearance and change of significant relations of the process. Such a representation makes it possible to distinguish processes that reach the same final state through different paths.

For example:

$$\text{process } A: \quad 0.1 \to 0.2 \to 0.8$$
$$\text{process } B: \quad 0.9 \to 0.8 \to 0.8$$

have the same final value 0.8, but the structures of their movement are different.

To preserve information about the sequence of changes, a *path signature* can be used. Conditionally, it has the form:

$$S(Y) = (1;\ \text{first level of movement};\ \text{second level of movement};\ \text{third level of movement};\ \ldots)$$

The first level reflects the total change. Subsequent levels make it possible to account for the order and interaction of changes in different coordinates of the trajectory. Therefore, the signature represents not only the question "How much did the process change?" but also "How did this change occur?"

Along with the signature, directly interpretable indicators can be used. For example, for a certain relation $v(\tau)$, let us introduce an indicator of early representation:

$$E = 2\int_0^1 (1-\tau)\, v(\tau)\, d\tau$$

and an indicator of late representation:

$$T = 2\int_0^1 \tau\, v(\tau)\, d\tau$$

In the first formula, earlier parts of the history receive greater weight; in the second, later parts receive greater weight. Therefore, two processes with the same average level of relation may have:

$$(E_1, T_1) \neq (E_2, T_2)$$

if the development of that relation occurred differently.

The joint movement of two relations can also be studied. For example:

$$\Gamma = \int_0^1 r(\tau)\, c(\tau)\, d\tau$$

where r and c are two different relations existing within one history. In this case, Γ characterizes their joint representation over time and is not interpreted as a statistical probability.

Thus, the original image of "flickering" receives a mathematical expression: what matters is not only the state of the process at a particular moment, but also the shape of the path by which the process arrived at that state.

7. Finding a Candidate for the Opposite. The mere presence of a process weakly related to P_0 is not sufficient to recognize it as an opposite. Such a process may appear by chance, may be poorly rooted in the development under consideration, or may differ from P_0 only locally. Therefore, a candidate undergoes several different tests.

Rootedness. The first type of test characterizes the extent to which the candidate arises from the entire observed system of relations. Let us denote this profile by R_i. What matters here is not only the direct relation between the candidate and P_0, but also its representation through other processes, their internal structures, and the history of their change. The candidate must be a part of the development of the original process rather than simply an external element that happened to appear among the generated results.

Structural Significance. The second type of test is associated with removing the candidate. Let us denote the corresponding indicator by I_i. Two versions of the same mathematical construction are compared: with candidate P_i and without candidate P_i. After the candidate is removed, the elements depending on it are recalculated. This makes it possible to test whether the presence of the candidate has a genuine structural effect on the discovered description of movement.

Independent Development. The third type of test is associated with further deepening of the candidate. Let us denote the result of this test by K_i. The candidate is temporarily treated as a local original process and is disclosed further. The original process P_0 remains the point of comparison. The test determines whether P_0 remains stably represented within the further development of the candidate or whether this representation ceases to be stable as the disclosure deepens.

Thus, a possible opposite is determined not by a single number but by a combination of three properties:

$$\text{rootedness} + \text{structural significance} + \text{independent development}$$

In other words: the candidate must arise from P_0, be significant for the discovered movement, and at the same time cease to stably preserve P_0 in its own development.

The exact decision functions, numerical thresholds, and technical configuration of these tests are not disclosed in the present article.

8. Why a Single Final Score Is Not Used. The obtained characteristics describe different properties of a candidate. Therefore, simply reducing them to a single number, for example:

$$F_i = \alpha R_i + \beta I_i + \gamma K_i$$

would introduce an additional assumption about the extent to which one indicator can compensate for another. For example, high rootedness could mathematically compensate for failure in independent development solely because the researcher selected particular coefficients α, β, and γ. Such an approach is not used in the proposed model.

Instead, the idea of a partial order is applied. The notation

$$i \succeq j$$

means that candidate i is no worse than candidate j according to all mandatory comparable criteria. Strict superiority

$$i \succ j$$

means that the first direction is confirmed while the reverse direction is not.

The set of non-dominated candidates is then defined as the set of processes for which there is no other candidate that dominates them according to the adopted set of criteria. Conditionally:

$$O = \{\, i : \text{there is no } j \text{ for which } j \succ i \,\}$$

Thus, the algorithm does not have to terminate by selecting a single winner. The result may be: several non-dominated candidates; incomparable candidates; a single candidate; or insufficient data for a conclusion. This corresponds to the original formulation of the problem: each finite iteration represents only a finite disclosure of the process, not its entire potentially infinite picture.

9. Boundaries of the Mathematical Model. The proposed formalization has several fundamental limitations.

First, the function s is an external semantic evaluator. The mathematical model uses the values it produces, but does not itself prove the truth of the corresponding semantic relation.

Second, the operator H describes a common structure of observed transitions. Even the successful construction of such an operator does not yet prove the existence of a universal law of development outside the investigated domain.

Third, any finite characteristics of a trajectory inevitably lose part of the information contained in the complete path. Therefore, the original history of states must be preserved alongside its mathematical representations.

Fourth, the model always operates with a finite disclosure of a potentially infinite process. Therefore, the absence of a certain opposite among the examined processes means only "it was not found within the given finite configuration and under the given disclosure procedure" and not "such an opposite does not exist."

The present work deliberately does not provide the complete dictionary of functions of the operator H, the exact rules governing the formation of its permissible inputs, the technical configuration of the semantic evaluator, or the full final selection procedure. These elements belong to the further formalization and experimental implementation of the method.

The mathematical formalization of the oscillatory character of development, introduced earlier at the level of the author's intuition, also remains separate. The mere fact that the number of processes may increase or decrease from one iteration to another is not yet a mathematical definition of oscillation. Therefore, a separate oscillatory term is not yet introduced into the present mathematical model.

As a result, the proposed construction translates the original idea of development into a sequence of finite addressed states:

$$Z_0 \ \to \ Z_1 \ \to \ Z_2 \ \to \ \cdots \ \to \ Z_t \ \to \ \cdots$$

The history of these states is used to search for a common transition structure H. The form of development of individual processes is then analyzed as a trajectory, while candidates for the opposite are tested from several independent perspectives.

Thus, the subject of the mathematical description is not only the state of the system under consideration, but above all **the structure of its change**.

Conclusion

The world can be described dialectically — not only through the laws of physics or probability, but also through movement. The proposed algorithm is an attempt at such a description. Much work remains to be done in this direction: from refining the propositions of the algorithm itself, through bringing the mathematical scaffolding to completeness, to the program code with which one can test what the algorithm actually yields in practice.